

Algebraic Topology: a beginner's course

This lecture:

AlgTop23: Knots and surfaces II

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Lecture 23: Knots & surfaces II



boundary.



Steps:

1. Orient knot



2. Trace knot, at each crossing change to other strand, still in same direction



Continue

3. Create disjoint disks with these
Seifert circles as boundaries



4. Join disks
crossing
bands, ea
previous
and rectangular
ball

links with these
as boundaries



at previous
twisted rectangular
+ twist, following crossing

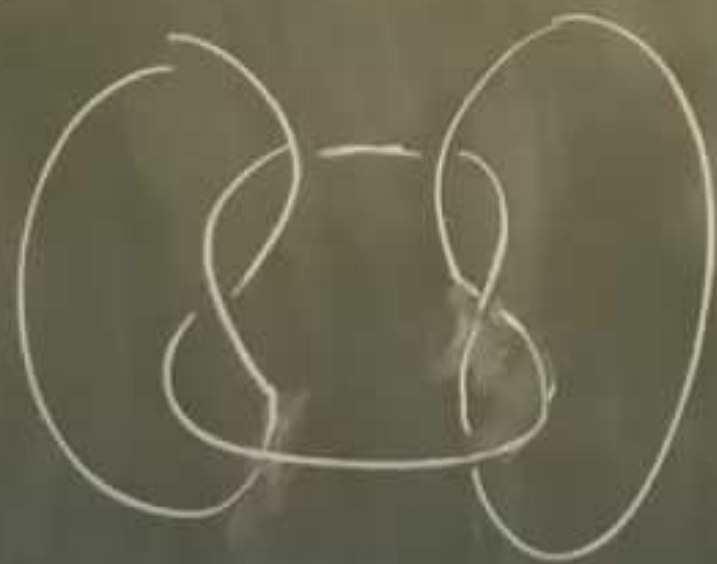


a Seifert
surface
(of knot)

boundary is original trefoil
+ surface is orientable

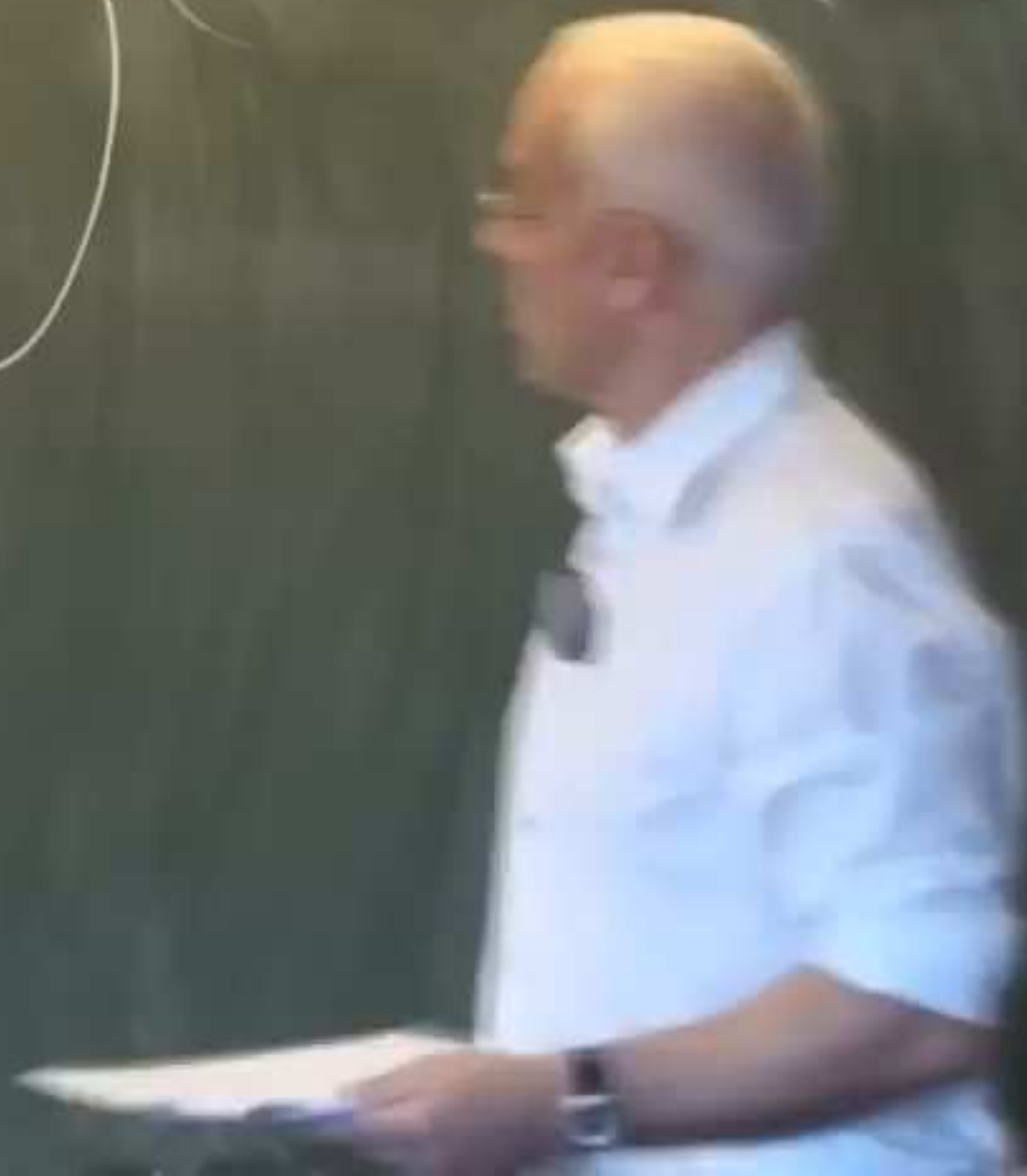
Ex Square knot:

Seifert
surface
(of knot)



original trefoil

orientable

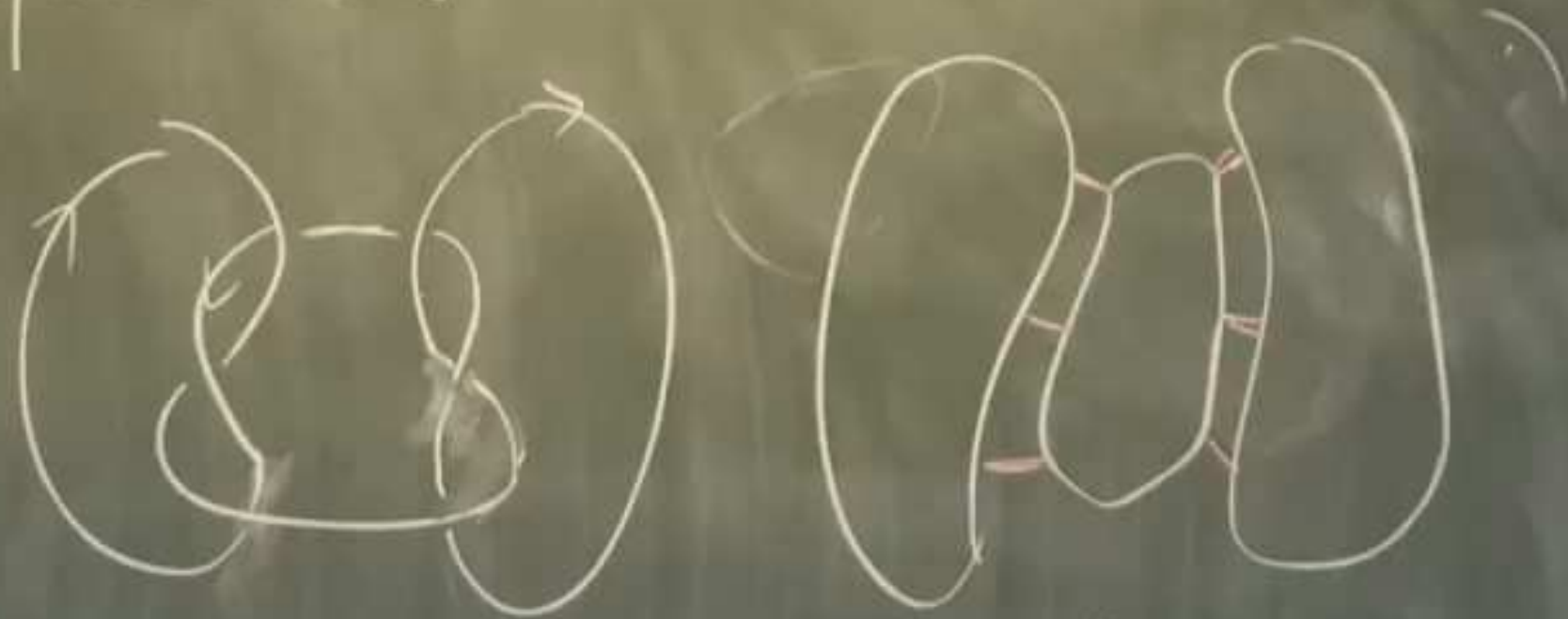


Ex Square knot

Seifert
surface
(of knot)

original trefoil

orientable



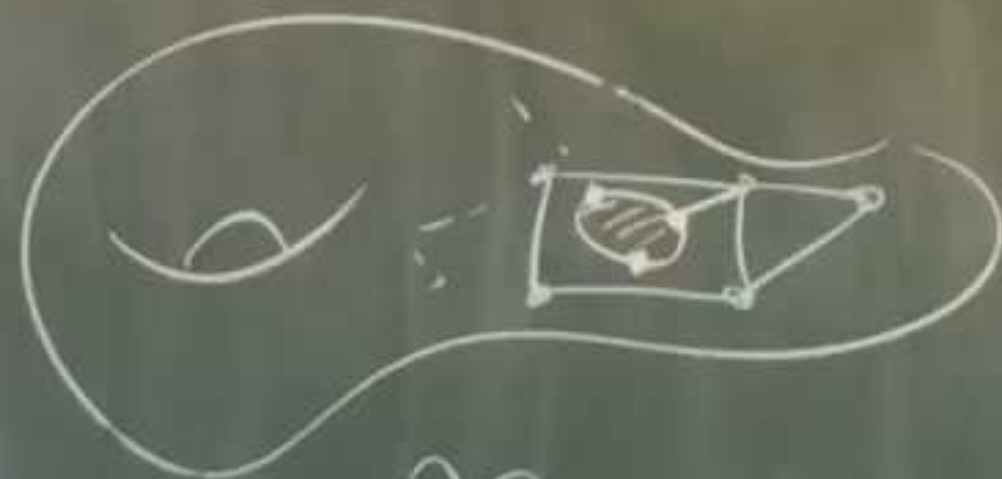
3 disks, 6 bands

closed disk)

cuff
= χD

Q. What happens to Euler number χ if you add a cuff?

Ex Square knot



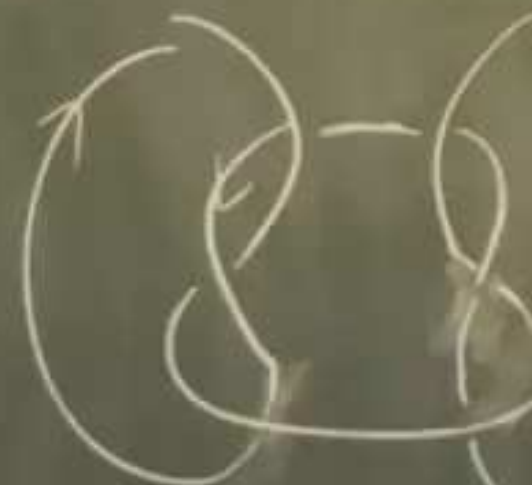
$$V \rightarrow V$$

$$E \rightarrow E$$

$$F \rightarrow F$$

$$m \quad \chi$$

$$\boxed{\chi \rightarrow \chi}$$



3 di

es

happens to Euler number
you add a cuff?



$$V \rightarrow V+3$$

$$E \rightarrow E+3+1$$

$$F \rightarrow F$$

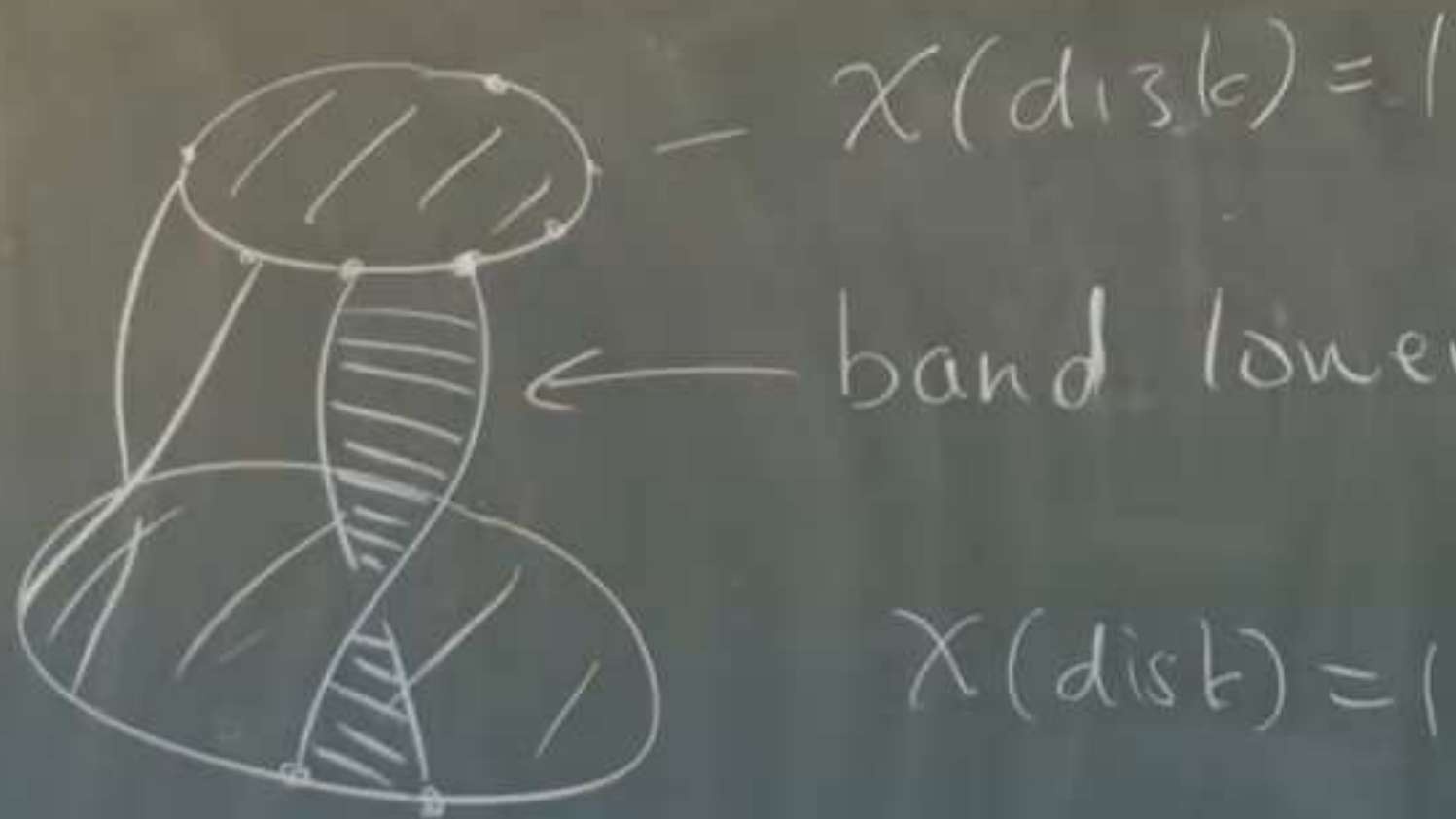
$$\chi \rightarrow \chi - 1$$

each cuff reduces
 χ by 1

Siefert
surfaces

- sphere with k cuffs $\chi = 2 - k$
- n -holed torus with k cuffs $\chi = 2 - 2n - k$
- n -crosscap with k cuffs $\chi = 2 - n - k$

Seifert surface. d disks + b bands



- trefoil. 2 disks, 3 bands $X(\text{Seifert surface})$
 $= 2 - 3 = -1$

e - d disks + b bands

$$\chi(\text{disk}) = 1$$

band lowers χ by 1

$$\chi(\text{disk}) = 1$$

ts, 3 bands $\chi(\text{Seifert})$

$$= 2 - 3$$

so S.S must have

$$\chi = 2 - 2n - k \text{ where}$$

$$= -1$$

$$2 -$$